Renewable Energy 83 (2015) 646–657

Contents lists available at ScienceDirect

Renewable Energy

journal homepage: www.elsevier.com/locate/renene

Integrated sizing and scheduling of wind/PV/diesel/battery isolated systems

André Malheiro [a, *], Pedro M. Castro [b], Ricardo M. Lima [c], Ana Estanqueiro [a, d]

[a] Departamento de Engenharia Geográfica, Geofísica e Energia, Faculdade de Ciências, Universidade de Lisboa, 1749-016 Lisboa, Portugal
[b] Centro de Investigação Operacional, Faculdade de Ciências, Universidade de Lisboa, 1749-016 Lisboa, Portugal
[c] Computer, Electrical and Mathematical Sciences and Engineering Division, King Abdullah University of Science and Technology (KAUST), Thuwal 23950-6900, Saudi Arabia
[d] Unidade de Análise Energética e Redes, Laboratório Nacional de Energia e Geologia, 1649-038 Lisboa, Portugal

ARTICLE INFO

Article history:
Received 14 June 2014
Accepted 28 April 2015
Available online 22 May 2015

Keywords:
LCOE
Isolated hybrid systems
Sizing optimization
Renewable energies

ABSTRACT

In this paper we address the optimal sizing and scheduling of isolated hybrid systems using an optimization framework. The hybrid system features wind and photovoltaic conversion systems, batteries and diesel backup generators to supply electricity demand. A Mixed-Integer Linear Programming formulation is used to model system behavior over a time horizon of one year, considering hourly changes in both the availability of renewable resources and energy demand. The optimal solution is achieved with respect to the minimization of the levelized cost of energy (LCOE) over a lifetime of 20 years. Results for a case study show that the most economical solution features all four postulated subsystems.

© 2015 Elsevier Ltd. All rights reserved.

1. Introduction

The world energy demand is increasing at a larger rate than the growth of human population [1]. It is expected that the highest increase will occur in the developing countries, where 1.4 billion people still do not have access to energy services [2]. The lack of an interconnected electrical grid and synchronous power system in remote areas might be overcome by the installation of isolated micro-grids (electric islands) fed by endogenous resources. These type of isolated systems have been implemented using only diesel generators or combined with one renewable source, but due to the growing maturity of wind and photovoltaic systems, and high diesel costs, these traditional systems are being updated in order to incorporate multiple renewable technologies [3] [4].

The systematic design and operation of hybrid systems by optimization tools is driven by the need to overcome the complexity of decision making in the presence of high variability of renewable resources, variable energy demand profile, different performance and cost characteristics of equipment.

The evaluation of hybrid systems has been reported using different performance models, optimization software tools, and techniques [5] [6]. The methodology proposed in Ref. [7] uses a dynamic programming model to determine the optimal operating strategy for a wind-diesel-battery system during 24 h. The design optimization of a Wind/Diesel/Battery system based on construction of trade-offs curves is presented in Ref. [8] so as to define the diesel and battery system capacity. In addition, the economics of charging the batteries with diesel power is discussed comparing alternative models for dispatch strategies. In Ref. [9], the authors propose a non-linear constrained model to find the optimal combination of energy components (hydro, wind, solar, diesel and batteries) using an iterative method and a Quasi-Newton algorithm. Several individual combinations are compared, and the best is chosen. A sizing method for an isolated grid with diesel is presented in Refs. [10] and [11]. The authors have used an optimization framework based on genetic algorithms to select the optimal wind turbine, PV rated power, nominal capacity for the battery energy storage system, and inverter rating. The benefits of introducing renewables compared to a 100% diesel system are then measured. However, in both works, the diesel generators are outdated and the re-size is not evaluated for an optimal solution. A similar approach is followed in Ref. [12] but without the diesel backup. A methodology based on a multi-objective

* Corresponding author. Tel.: +351 969 855 242.
E-mail address: andrefmalheiro@gmail.com (A. Malheiro).

http://dx.doi.org/10.1016/j.renene.2015.04.066
0960-1481/© 2015 Elsevier Ltd. All rights reserved.

evolutionary algorithm is proposed in Ref. [13] to optimize the system cost and the CO_2 emissions. One simulation tool widely used is the Hybrid Optimization Model for Electric Renewables (HOMER) [14]. However, this tool does not automatically optimize the design of the system, being necessary to pre-set the size of the components.

Other recent works have focused on mixed-integer linear programming (MILP) formulations to optimize micro-grids, being the Microgrid group at the Lawrence Berkeley National Laboratory (LBNL) in California, USA, one of the top contributors. The optimal scheduling of hybrid systems is discussed in Ref. [15] whereas [16] addresses an energy management system.

In some cases, diesel generators are not considered for system evaluation, leading to the installation of large storage capacity to cope with a system relying 100% on renewables. Naturally, this has a strong impact on overall system cost.

From this literature review it is clear that different sub-systems have been studied, different types of optimization methods have been used, and different time horizons considered. More specifically, some authors have focused on specific sub-sets of fixed subsystems of equipment, for example a system with only diesel and wind [7], while others have considered only renewable sources, or multiple systems [14]. Some authors prefer to apply stochastic-based search algorithms [12], while lately there is a clear trend to use MILP models. The advantage of working with an MILP model is that one can easily compute a best possible solution, which is valuable information whenever optimality cannot be proved. Furthermore, there have been considerable improvements at the level of MILP solvers [17] [18], which allow obtaining optimal or near optimal solutions in short CPU times with modern day computers. The length of the time horizon is also important. If one is concerned with the optimal operation for a fixed design, a short-term horizon of 24 h might be enough. However, if the goal is to design the system, longer time horizons should be considered so as to account for seasonal changes in weather data and obtain a more robust solution.

The main contribution of this work is to provide an integrated methodology for the sizing and scheduling of hybrid systems involving the main renewable sources, solar and wind power, batteries for storage and diesel generators as a backup system. We rely on mathematical programming as the solution technique and propose a mixed-integer linear programming model that is capable of: a) choosing which subsystems are present in the optimal configuration; b) determining the optimal sizing of equipment units considering long-term operation.

The rest of the paper is organized as follows. The problem definition is given in section 2, followed by the model of the hybrid system in section 3. Section 4 is concerned with the computation of the levelized cost of energy as a function of the key design variables, while section 5 defines the optimization problem and lists the software tools that were used to solve it. The problem data for our case study is given in section 6, with the results being discussed in section 7. A sensitivity analysis is then performed in section 8, before the conclusions in section 9.

2. Problem definition

Four possible subsystems are considered for the autonomous hybrid system: wind energy, photovoltaic, batteries, and diesel generators. Assumed given are the availability of the endogenous renewable resources, energy demand profile, as well as the general technical characteristics of commercially available equipment (rated power, efficiency, and capital cost). The objective is the optimal design of the system that minimizes the levelized cost of energy (LCOE), while meeting energy demand under uninterruptible conditions. More specifically, we aim to determine the number of wind turbines, PV panels, diesel generators, and batteries, the latter providing the required storage capacity to allow uninterrupted operation. Note that batteries and diesel have complementary strengths as a backup for renewables. Batteries are extremely expensive and their lifespan is dependent on the number of charge/discharge cycles, and so their size and usage should be kept to a minimum. In contrast, diesel generators are inexpensive, easy to use and have a longer lifespan, but because fuel consumption is high when operating in transient modes - something the batteries respond very well to – their use should be reserved for long periods of energy deficit. The time horizon considered is one year with a time step discretization of one hour.

2.1. Remarks

From the modeling point of view, it is straightforward to consider a smaller time step discretization but that would significantly increase the number of variables and constraints leading to a larger computational time. In addition, the proposed methodology is intended as a planning tool for a time horizon of one year, which makes the chosen discretization time step adequate considering the intraday variability of resources and energy demand. In contrast, operational dynamic tools to be used in the transient or quasi-stationary analysis of power systems (e.g. for power quality assessment or load flow simulation) may need to consider shorter time horizons. However, their range of application is usually quite specific and limited by the complexity of the underlying phenomena and corresponding models of subsystems and therefore are not adequate for the design and optimal sizing of hybrid systems.

3. Hybrid system modeling

In the proposed model, most subsystems are represented as a number of units, to be determined by the optimization, of a specific type and size (e.g. two wind turbines of 20 kW, six hundred PV panels of 0.25 kW, one diesel generator of 17.6 kW), while the battery storage capacity is considered as a single equivalent battery. The auxiliary equipment (e.g. inverters) is included in the efficiency and capital cost of the main equipment, and therefore, their size and numbers are not determined. The energy surplus is dissipated in electric resistances. It is also assumed that the power generated by each source, and the charge/discharge power of batteries, is constant during the time step discretization. Therefore, the power is numerically equal to the energy in each time period of 1 h, for the common units used in the power sector, i.e., kW and kWh.

3.1. Wind energy subsystem

The power curve of a wind turbine gives the electrical power output as a function of wind velocity. The characteristic curve from the generic wind turbine considered in this work is depicted in Fig. 2 and was obtained from averaging normalized power curves of commercial wind turbines (power curves adimensionalized by the rated power of each turbine). More specifically, the energy (kWh) produced by a wind turbine in time step t is given by:

$$E_{wt}(t) = P_{WT} \times \left[a_1 \times e^{-\left(\frac{u_z(t)-b_1}{c_1}\right)^2} + a_2 \times e^{-\left(\frac{u_z(t)-b_2}{c_2}\right)^2} + a_3 \times e^{-\left(\frac{u_z(t)-b_3}{c_3}\right)^2} \right] \times \Delta t \quad \forall t, \tag{1}$$

where P_{WT} is the rated power (kW) and a_1 to c_3 are the coefficients of the regression. From the curve in Fig. 2, it is easy to check that the efficiency of the wind turbine (measured in p.u./(ms^{-1})) is not constant but varies with the wind velocity.

The electrical energy generated is limited by the range of operation of the wind turbine, between the cut-in and cut-out wind speed. Considering the hub installation height of the wind turbine, the wind speed data is vertically transposed by using a power law:

$$u_z(t) = u_{zREF}(t) \times \left(\frac{z}{z_{REF}}\right)^{\frac{1}{n}} \quad \forall t, \tag{2}$$

where $u_z(t)$ is the wind profile (m/s) for rotor installation height (m) z, $u_{zREF}(t)$ is the average measured wind speed (m/s) at z_{REF} height (m) in the time step t, and n is the surface roughness.

3.2. Photovoltaic energy subsystem

The energy generated by a PV panel in time step t is determined as a function of the cell temperature, and the solar irradiance [22]:

$$E_{PV}(t) = PMP_{REF} \times \frac{G(t)}{1000\ W/m^2}[1 + \gamma_{PV} \times (T_{cell}(t) - 25°C)] \times \eta_{DA} \times \Delta t \quad \forall t, \tag{3}$$

$$T_{cell}(t) = T_{amb}(t) + \frac{NOCT - 20°C}{800\ W/m^2} \times G(t) \quad \forall t, \tag{4}$$

where PMP_{REF} is the point of maximum power (kW) of a PV panel in Standard Test Conditions (STC, 1000 W/m^2, and 25 °C cell temperature), $G(t)$ is the solar irradiance (W/m^2) measured in the panel inclination installation (equal to the site latitude), γ_{PV} is the temperature power coefficient (%/°C), η_{DA} is the efficiency (%) of power converter, $T_{cell}(t)$ is the cell temperature (°C), $T_{amb}(t)$ is the measured ambient temperature (°C), and $NOCT$ is the nominal operation cell temperature (°C). PMP_{REF}, $NOCT$, and γ_{PV} are provided by the panel manufacturer.

3.3. Diesel generator subsystem

Similarly to other works [13], the diesel consumption C_{dg} (L) of the generator system is assumed to be given by a linear function:

$$C_{dg}(t) = d_1 \times E_{dg}(t) + d_2 \times P_{dg} \times n_{dg}(t) \times \Delta t + nA(t) \times pA \ \forall t, \tag{5}$$

where d_1, and d_2 are coefficients of specific diesel consumption (L/kWh) obtained from commercial equipment data [23] [24] [25], $E_{dg}(t)$ is the energy (kWh) produced in the time step t by this subsystem, P_{dg} is the rated power (kW) of one diesel generator, $n_{dg}(t)$ is the number of working generators, $nA(t)$ is the number of generator starts and pA is the diesel startup penalty (in L) [8]. It should be noted from Eq. (5) that the generation efficiency (kWh/L) is not constant, it increases with the power output level $E_{dg}(t)$ as the relative importance of the second term on the right-hand side decreases (P_{dg} is a parameter, not a variable).

To avoid operation with low efficiency, a minimum service level DG_{min} is enforced (e.g. 30%):

$$E_{dg}(t) \geq DG_{min} \times P_{dg} \times y(t) \times \Delta t \quad \forall t, \tag{6}$$

where $y(t)$ denotes a binary variable that is equal to 1 if the generator is working, thus guaranteeing that it is always generating above the lower bound. The number of generators to install, n_{dgt}, is then determined by the maximum working generators, whereas the annual working hours, n_{dgh}, are incremented in every hour they are working.

3.4. Batteries subsystem

The batteries act as an energy storage subsystem that is modeled as a single equivalent battery. Its performance is characterized by the efficiency of charging and discharge modes according to Fig. 1 and [26]. The energy balance defines the batteries state of charge, E_{bat}, in kWh:

$$E_{bat}(t) = E_{bat}(t-1) + E_{bin}(t) \times (\eta_{BC} \times \eta_{AD}) - \frac{E_{bout}(t)}{\eta_{BD} \times \eta_{DA}} \ \forall t, \tag{7}$$

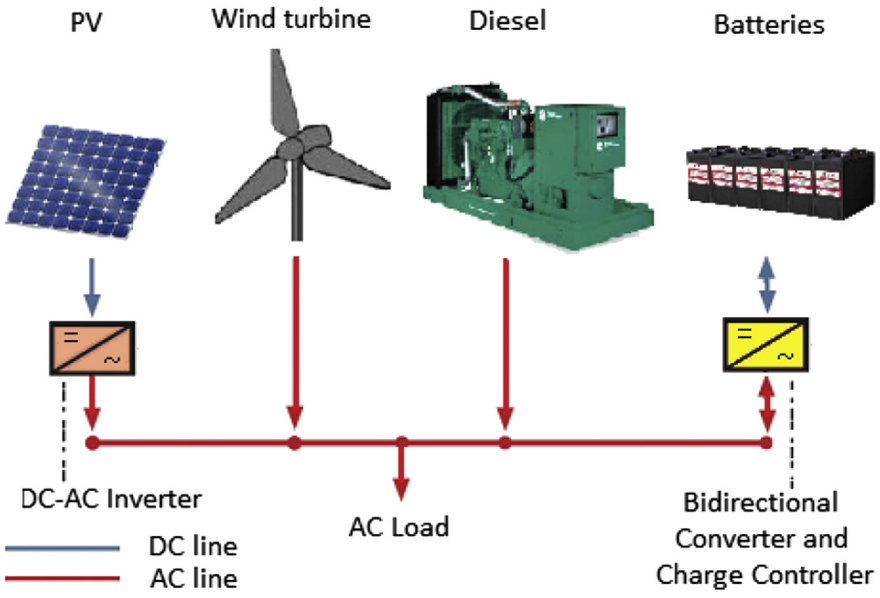


Fig. 1. AC microgrid configuration.

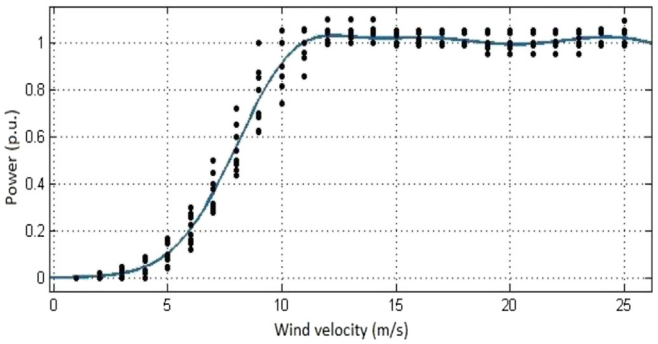


Fig. 2. General wind turbine power curve. The dots represent the normalized power curves from commercial equipment [19] [20] [21].

where $E_{bout}(t)$, and $E_{bin}(t)$ respectively represent the energy (kWh) used from the batteries, and the energy available to charge the batteries in the time step t, η_{BC} and η_{BD} are the battery charge and discharge efficiency (%), while η_{AD} and η_{DA} are power equipment efficiency (%) in AC-DC and DC-AC conversions. The storage capacity to install, C_{bat}, will be the maximum amount of stored energy over all time steps t.

In each time period, it is guaranteed a minimum energy level in the batteries and a maximum amount of energy battery flow:

$$E_{bat}(t) \geq C_{min} \times C_{bat} \quad \forall t, \tag{8}$$

$$E_{bin}(t) \times (\eta_{BC} \times \eta_{AD}) + \frac{E_{bout}(t)}{\eta_{BD} \times \eta_{DA}} \leq F_{max} \times C_{bat} \times \Delta t \quad \forall t, \tag{9}$$

$$E_{bin}(t) \leq C_{bat} \times \delta(t) \quad \forall t, \tag{10}$$

$$E_{bout}(t) \leq C_{bat} \times \varphi(t) \quad \forall t, \tag{11}$$

$$\varphi(t) + \delta(t) \leq 1 \quad \forall t, \tag{12}$$

where C_{min} is the minimum state of charge (in %), F_{max} is the maximum charge/discharge rate (in %), $\delta(t)$ and $\varphi(t)$ are binary variables that register the battery energy flow. Note that in each time step, only one of the three following battery modes is possible: charge ($E_{bin}(t)$), discharge ($E_{bout}(t)$) or no flow of energy. In the beginning of the time horizon, the batteries are fully charged.

3.5. Integrated hybrid system

The balance between the amount of energy that is used (demand plus batteries charge) and the available energy (renewables, diesel, and batteries discharge) is verified according to:

$$E_{dem}(t) + E_{bin}(t) = Eu_{WT}(t) + Eu_{PV}(t) + E_{dg}(t) + E_{bout}(t) \quad \forall t, \tag{13}$$

where $E_{dem}(t)$ is the energy demand in time step t, in kWh, $E_{bin}(t)$ is the energy to charge the batteries, $Eu_{WT}(t)$, and $Eu_{PV}(t)$ are the used energy from the wind, and PV subsystem, respectively, $E_{dg}(t)$ is the energy produced by the diesel generators subsystem, and $E_{bout}(t)$ is the energy used from the batteries.

A similar balance is established for the evaluation of the energy surplus. It is assumed that only the renewable sources contribute to the surplus, which is described by the following equations:

$$n_{WT} \times E_{WT}(t) = Eu_{WT}(t) + Es_{WT}(t) \quad \forall t, \tag{14}$$

$$n_{PV} \times E_{PV}(t) = Eu_{PV}(t) + Es_{PV}(t) \quad \forall t, \tag{15}$$

where n_{WT}, and n_{PV} are the number of wind turbines, and PV panels to install, $E_{WT}(t)$, and $E_{PV}(t)$ are the energy produced by single units described by Eqs. (1) and (3), and $Es_{WT}(t)$, and $Es_{PV}(t)$ denote the energy surplus from each source.

4. Levelized cost of energy

The LCOE, expressed in €/kWh, is the criterion used to rank the different designs of hybrid isolated systems. The LCOE provides a methodology to compare total costs for different time periods and between different systems and encompasses the initial capital, discount rate, operation, maintenance, fuel, and replacement costs. In this work, it is assumed that the annual energy produced and the annual costs will be constant during project lifetime. When this kind of assumptions is made, it is possible to determine a capital recovery factor (CRF) that allows extrapolating results from one year to the entire lifetime of a project [27]. The total life cycle cost is determined for each system by:

$$LCOE = \frac{TLCC}{E_{con}} \times CRF, \tag{16}$$

$$TLCC = I_0 + \frac{O\&M}{CRF} + \frac{C_R}{CRF}, \tag{17}$$

$$CRF = \frac{d \times (1+d)^{A}}{(1+d)^{A} - 1}, \tag{18}$$

where $TLCC$ is the total life cycle cost (€), E_{con} is the annual energy demand (kWh), CRF is the capital recovery factor (%), I_0 is the capital cost (€), $O\&M$ is the annual cost (€) of operation and maintenance, C_R is the annual replacement cost (€), d is the nominal discount rate (%), and A is the useful lifetime of the project in years [27] [28].

The minimum LCOE over the lifetime of the project is obtained by minimizing the TLCC subject to the operating constraints of the four subsystems, assuming that E_{con}, $O\&M$, and C_R are constant during the project years. Then, the system is optimized for one year, and the results extrapolated for the useful lifetime. The objective function is determined by:

$$\text{Minimize} : TLCC = TLCC_{WT} + TLCC_{PV} + TLCC_{dg} + TLCC_{bat}, \tag{19}$$

where $TLCC_{WT}$, $TLCC_{PV}$, $TLCC_{dg}$, and $TLCC_{bat}$ are respectively the total life cycle cost (€) of wind, PV, diesel generators and batteries subsystems.

$TLCC_{WT}$ and $TLCC_{PV}$ are given by the capital cost of equipment (I_{WT}, and I_{PV}) in €/kW [27] [28]:

$$TLCC_{WT} = n_{WT} \times P_{WT} \times I_{WT} \times \left(1 + \frac{O\&M_{WT}}{CRF}\right), \tag{20}$$

$$TLCC_{PV} = n_{PV} \times PMP_{REF} \times I_{PV} \times \left(1 + \frac{O\&M_{PV}}{CRF}\right). \tag{21}$$

For each subsystem, the annual operation and maintenance costs ($O\&M_{WT}$, $O\&M_{PV}$) are given as percentages (%) of the capital cost. The PV capital cost already includes the PV inverter and other auxiliary equipment capital and replacement cost.

The TLCC of diesel generators is given by capital cost (I_{dg}) in €/kW, the maintenance cost (M_{dg}) in €/h, and operation cost (O_{dg}) in €/L [27]:

$$TLCC_{dg} = n_{dg_t} \times P_{dg} \times I_{dg} + \frac{n_{dg_h} \times M_{dg} + \sum_{t=1}^{T} C_{dg} \times O_{dg}}{CRF}, \quad (22)$$

where n_{dgt} is the number of diesel generators to install, P_{dg} is the rated power in kW of one generator, and n_{dgh} is the annual working hours. The TLCC of batteries include the capital cost of batteries, and power equipment (I_{bat}, and I_{PE}) in €/kWh, and €/kW, respectively, annual operating cost (O_{bat}) in €/kWh, annual maintenance cost (M_{bat}) in €/kW, and annual replacement cost in €/kWh (batteries and batteries inverters - C_R, one time replacement during project lifetime, but costs are reported as an annual cost) [27] [29]:

$$TLCC_{bat} = C_{bat} \times (I_{bat} + I_{PE} \times F_{max}) + \frac{O_{bat} \times \sum_{t=1}^{T} (E_{bout_e}(t) + E_{bin_e}(t))}{CRF} + \frac{M_{bat} \times F_{max} \times C_{bat} + C_R \times C_{bat}}{CRF}, \quad (23)$$

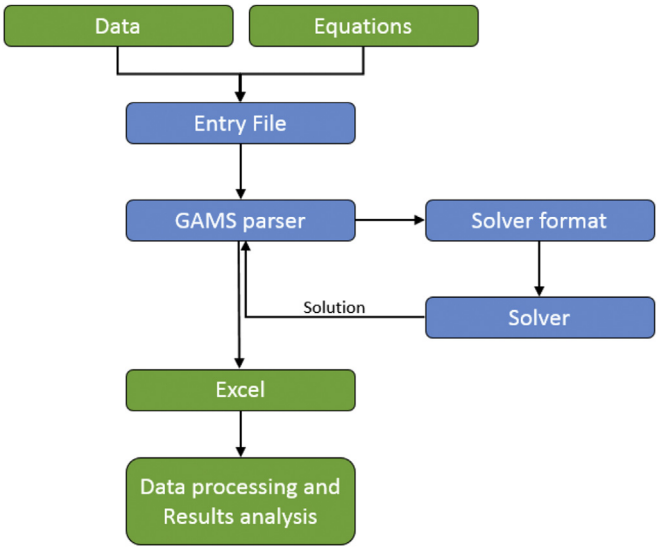


Fig. 3. Flowchart of the optimization approach where the GAMS software is the central element, handling data input and output, besides the solution of the optimization problem by one of its solvers.

where $E_{bine}(t)$, and $E_{boute}(t)$ are the energy (in kWh) that effectively charge, and discharge the batteries.

5. Optimization problem

The mixed-integer linear programming (MILP) optimization problem consists of the objective function in Eq. (19) subject to constraints of Eqs. (5)–(15), and (20)–(23). The number of wind turbines, PV panels, and diesel generators are defined as integer variables, the storage capacity is a continuous variable, and binary variables define logic relations between subsystems operating modes (e.g. batteries either charge or discharge). Note that working with the number of equipment units of a specific type instead of an aggregate size, avoids the post-processing step of rounding sizes to match existing commercial equipment, which typically increases cost. It also corresponds to a more modular design.

The optimization model is implemented in GAMS and is able to determine the design and operation of the hybrid system for a specific scenario characterized by given time-varying renewable resources availability and demand profiles, as well as equipment cost data. After the optimal solution is found, the values of the optimal variables are exported to Excel for detailed analysis and generation of various resource profiles. The flowchart of the proposed methodology is illustrated in Fig. 3. All optimization runs were performed using GAMS/CPLEX 12.5.1 as the MILP solver using 2 threads for parallel deterministic optimization. The stopping criteria was either a maximum CPU time of 10,800 s or an optimality gap of 5%, and the hardware consisted of an Intel Core2 2.4 GHz, 64 bits CPU, and 3.37 Gb of RAM.

6. Case study

We aim to design an isolated hybrid system for supplying the energy required by an industrial facility that starts working at 8 AM and stops at 7 PM. Demand is assumed to be equal for every day of the year, which is typical of many industries operating intensively in countries without interconnected power systems. The maximum hourly energy demand is 72 kWh, the daily consumption is 793 kWh, and the annual is 289,445 kWh. This demand profile is labeled Load A in Fig. 4.

Concerning the availability of renewable resources, we rely on meteorological data from the software SolTerm [30]. More specifically, we assume a synthetic hourly wind speed representative of Portugal (Fig. 5), and a 35° inclination hourly irradiance of Lisbon (Fig. 6).

The technical and economical parameters characterizing the different equipment are provided in Tables 1–6.

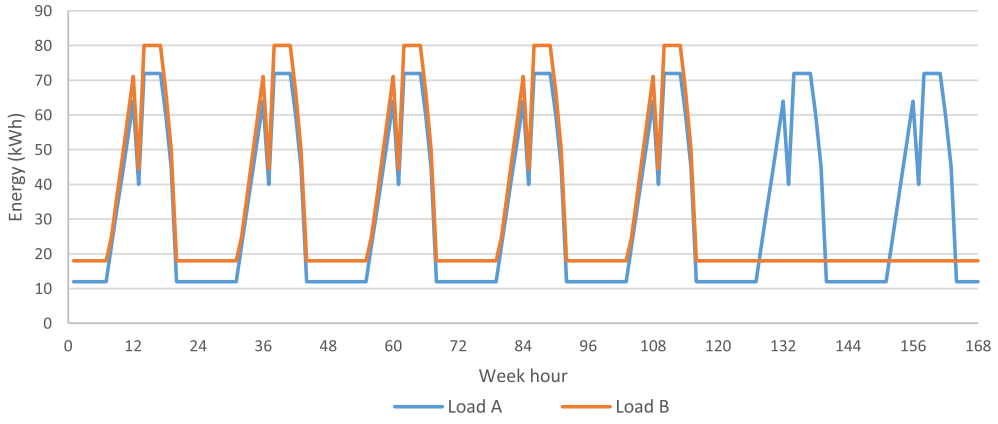


Fig. 4. Weekly energy demand profiles. Load A is used in all scenarios with exception of the sensitivity analysis of the demand profile, where Load B is considered.

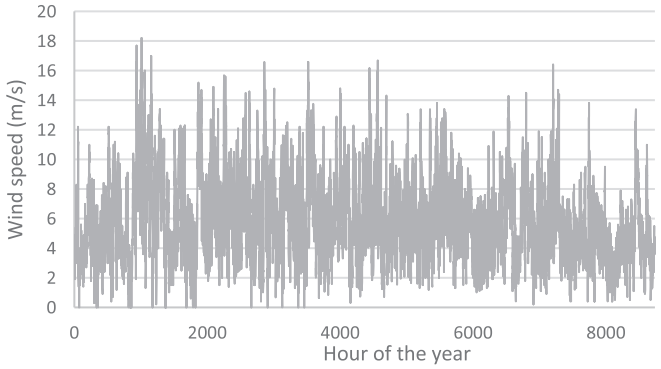


Fig. 5. Wind speed hourly data for one year. Average wind speed of 5.93 m/s at 40 m high.

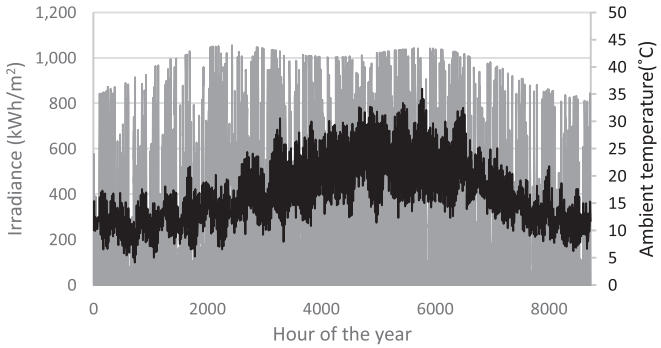


Fig. 6. Irradiance and temperature hourly data for one year. Total annual of 1872 kWh/m^2 for an angle of 35°, and average ambient temperature of 16.3 °C.

Table 1
Financial parameters.

	Parameter	Value	Unit
d	Discount rate	6	%
A	Useful life time	20	years

Table 2
Wind turbine parameters.

	Parameter	Value	Unit
P_{WT}	Rated power	20	kW
-	Cut in	2.5	m/s
-	Cut out	25	m/s
z	Rotor height	40	m
n	Roughness coefficient	–	–
I_{WT}	Initial investment	1784	€/kW
$O\&M_{WT}$	O&M cost	3	% Ini inv/year

Table 3
Photovoltaic parameters.

	Parameter	Value	Unit
PMP_{REF}	PMP panel	0.25	kW
γ_{PV}	Power temperature coefficient	−0.3	%/°C
$NOCT$	NOCT	41.5	°C
T_{REF}	Reference temperature	25	°C
I_{PV}	Initial investment	1520	€/kW
$O\&M_{PV}$	O&M cost	1	% Ini inv/year

Table 4
Diesel generator parameters.

	Parameter	Value	Unit
P_{dg}	Rated power	17.6	kW
DG_{min}	Minimum load	30	%
I_{dg}	Initial investment	333	€/kW
O_{dg}	Operation cost	1.421	€/L
M_{dg}	Maintenance cost	1	€/h
pA	Start penalty	0.7	L
$d1$	Fixed term for diesel consumption	0.3058	L/kWh
$d2$	Variable term for diesel consumption	0.0206	L/kWh

Table 5
Batteries parameters.

	Parameter	Value	Unit
C_{min}	Minimum state of charge	20	%
F_{max}	Maximum charge/discharge rate	8	%/h
η_{BD}	Discharge efficiency	100	%
η_{BC}	Charge efficiency	80	%
I_{bat}	Initial investment	213	€/kWh
M_{bat}	Maintenance cost	9.80	€/kW-year
O_{bat}	Operation cost	0.0004	€/kWh

Table 6
Power equipment parameters.

	Parameter	Value	Unit
η_{AD}	Efficiency AC-DC	93.4	%
η_{DA}	Efficiency DC-AC	93.4	%
I_{PE}	Initial investment	190	€/kW

Table 7
Optimal sizing for reference scenario.

	Wind	PV	Diesel	Batteries
Equipment	2	646	2	–
Power (kW)	40	161.50	35.2	–
Storage (kWh)	–	–	–	558.4
TLCC (€)	95,915	273,636	185,456	184,151
LCOE (€/kWh)	0.2232			

7. Results

Table 7 gives the optimal values of the key decision variables, leading to a total system cost of 739,158 € (sum of each TLCC subsystem) and an LCOE = 0.2232 €/kWh. Notice that the PV subsystem is the main component of the hybrid system in terms of installed capacity and cost. This result is not particularly surprising given the high correlation between the demand in Fig. 4 and the solar radiation profiles. As a consequence, the storage capacity provided by the batteries is relatively low for an isolated system. In turn, the diesel generators are used in periods with little or no PV production, typically during winter evenings (Fig. 7).

Diesel generation is much less frequent during spring and summer as can be seen in Fig. 8 and Fig. 9. It is interesting to see that sometimes (e.g. circle in Fig. 8) solar energy is enough to meet the total demand. This does not mean that there is no wind power available, simply that the model does not differentiate between the two subsystems (from an economic point of view, it is indifferent which subsystem is supplying energy), see Eqs. (13), (14), and (15). In other words, the total wind production was considered surplus when generating the chart but it could be the other way around.

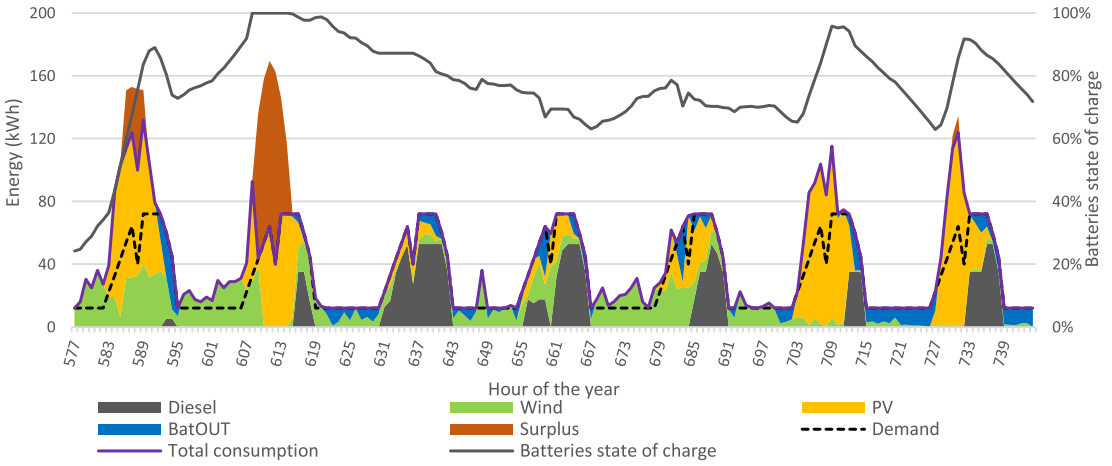


Fig. 7. Scheduling of consumption, production, and batteries state of charge (right y-axis) for the last week of January.

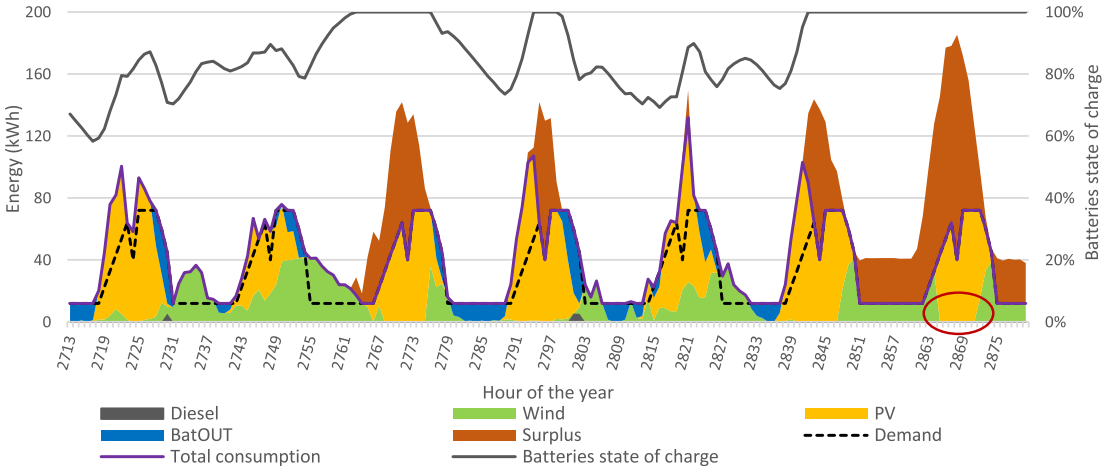


Fig. 8. Scheduling of consumption, production, and batteries state of charge (right y-axis) for the last week of April.

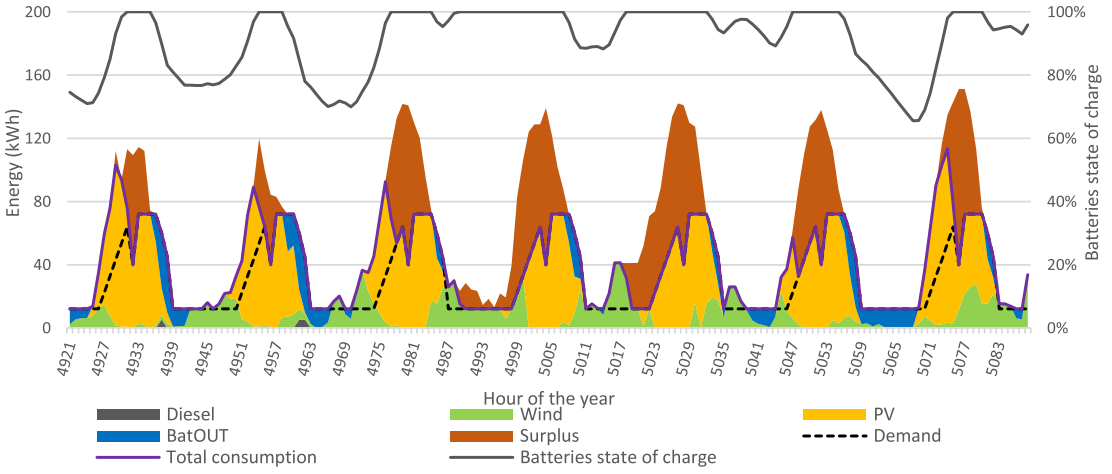


Fig. 9. Scheduling of consumption, production, and batteries state of charge (right y-axis) for the last week of July.

The total monthly balance of energy is presented in Fig. 10. Note that the solar production along the months is consistent with the solar resource availability, recall Fig. 6. November and December are the months with the lowest production of wind energy. The breakdown of the total energy production is as follows: 104,767 kWh (26.0%) from wind, 271,381 kWh (67.4%) from the PV subsystem, and 26,402 kWh (6.6%) from the diesel generators, corresponding to an annual diesel consumption of 9042 L. Of this,

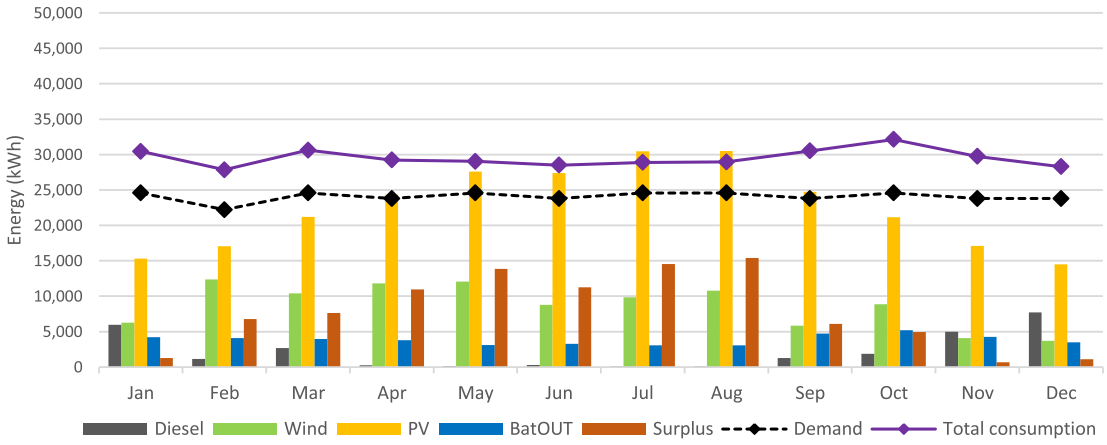


Fig. 10. Monthly production of each subsystem in one year for reference scenario.

23.5% is energy surplus and 4.9% are losses in converters so the actual renewable fraction accounts for 90.9%.

The batteries are an important mechanism to manage and balance the energy produced by the renewables, accounting for 16% of energy consumption. Considering the installed capacity of 558.4 kWh and the 20% minimum state of charge, their autonomy is 13.5 h at the average power demand of 33.0 kW. The diesel generators are sometimes also used to charge the batteries (this is due to the hard constraint on the generators minimum load).

The batteries begin the year in the maximum state of charge and finish in the minimum state allowed. Fig. 11 also shows that the state of charge of the batteries is generally high, over 60% most of the time except for very short periods. While it suggests poor utilization of the batteries, it should be noted that the objective of the optimization model is not their maximum utilization but the minimum capital and operating costs of the hybrid system during its lifespan, which is achieved by reducing the number of cycles (110/year), leading to roughly one replacement every 10 years (lead-acid batteries are, on average, replaced every 1000 cycles). Due to the high resource solar availability and installed PV capacity, the average state of charge is higher during the summer season.

8. Sensitivity analysis

A thorough sensitivity analysis is given next to evaluate decisions related to system configuration and to the test the impact of different economic, climate and demand data on system design. The different scenarios are summarized in Table 8.

8.1. System configuration

Alternative system configurations can easily be tested by forcing the installed capacities of specific subsystems to be equal to zero. The results in Table 9 show that the hybrid system with more components leads to the lowest LCOE, which was to be expected in the context of a rigorous optimization approach.

The Wind/PV/Diesel system without batteries has the same break-even point as the reference scenario but the installed capacity is the second highest, leading to a higher LCOE due to diesel consumption. The main difference between the PV/Diesel/Batteries and Wind/Diesel/Batteries systems lies on the renewable resource characteristics and the correlation between solar and demand profiles. Because of this, the operational costs with diesel in the Wind/Diesel/Batteries system are higher. The 100% Diesel system

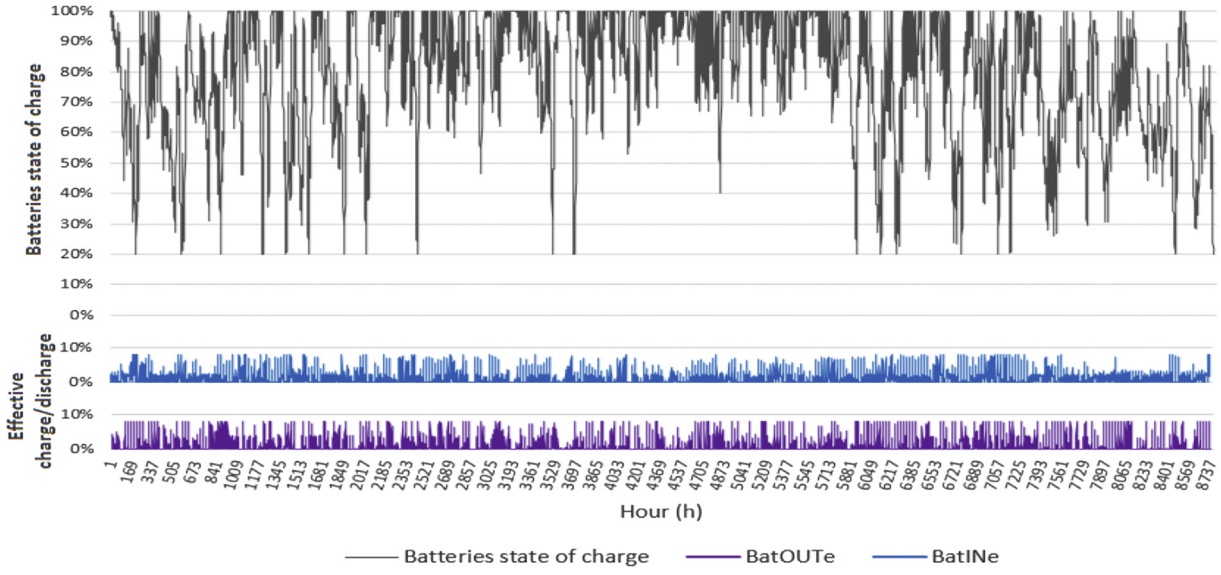


Fig. 11. Batteries state of charge over the year and energy that effectively gets in (BatINe) and out from batteries (BatOUTe).

Table 8
Working scenarios for sensitivity analysis.

Analysis	Scenario	Observations
System configuration	Reference scenario	Comparison between different hybrid configurations
Economic data	Diesel scenario	Diesel price variation
	Technology scenario	Technology cost variation (wind turbines, PV panels and batteries)
Climate data	Climate scenario 1	Excellent wind resource and poor solar resource
	Climate scenario 2	Average wind and solar resource
	Climate scenario 3	Poor wind resource and excellent solar resource
Demand profile	Load B instead of Load A	Same weekly demand but with higher/lower demand on weekdays/weekends

Table 9
Sizing results of reference scenario, and different configurations possibilities.

System	Installed capacity				Diesel (L)	LCOE (€/kWh)	Break-even point (years)
	Wind (kW)	PV (kW)	Diesel (kW)	Batteries (kWh)			
Wind/PV/Diesel/Batteries	40	161.50	52.8	558.4	180,844	0.2232	4.7
PV/Diesel/Batteries	–	228.50	35.2	865.3	160,434	0.2518	5.6
Wind/PV/Diesel	80	152.75	88	–	420,413	0.2664	4.7
Wind/Diesel/Batteries	160	–	52.8	991.8	394,900	0.3293	6.5
Wind/PV/Batteries	80	266.25	–	2026.3	–	0.3956	11.1
100% Diesel	–	–	88.0	–	1,941,338	0.5586	–

has the illusive advantage of leading to the lowest capital cost but it is the worst configuration considering the 20 years of the project lifespan due to the high operation costs, more than doubling the LCOE. Particularly interesting is the comparison of the base case to the Wind/PV/Batteries system. Notice that without the diesel backup system, the requested storage capacity of the batteries more than triples, the wind installed capacity doubles and the PV system capacity is higher.

8.2. Economic data

To evaluate the impact of equipment cost data on optimal system design, the following four parameters suffered a ± 50% change (one at a time): operational cost for diesel generators (O_{dg}), capital cost of wind generators (I_{WT}), PV panels (I_{PV}), and batteries (I_{bat}). The results in Fig. 12 show that the capital cost of PV is the main driver for LCOE variation, followed by diesel cost. On the other hand, the batteries have a reduced impact despite being the subsystem with the second largest capital cost. The variation in the renewable fraction does not exceed 6%, with a more competitive diesel price naturally leading to a higher energy production from fossil fuels.

8.3. Climate data

The weather scenarios represent three different combinations of renewable resources levels: (1) high wind and low solar; (2) average; (3) low wind and high solar. Ambient temperature data for computing the energy produced by the PV system was the same as in the reference case. The optimal designs are presented in Table 10.

In climate 1, we raise the average wind speed from 5.93 to 8.33 m/s and decrease the annual irradiance from 1872 to 1000 kWh/m^2. As a consequence, the system configuration changes drastically with almost all the energy coming from the wind turbines whereas the solar output becomes residual (Fig. 13). Diesel

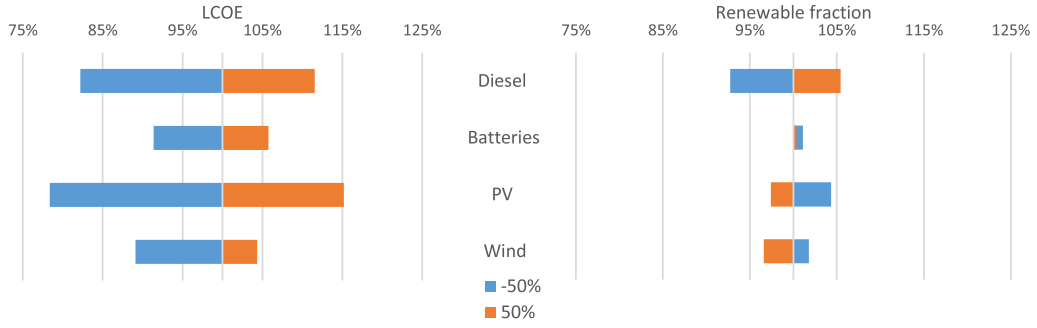


Fig. 12. LCOE and renewable fraction change with economic data.

Table 10
Sizing, and economical results for climate scenario 1, 2, and 3.

Scenario	Installed capacity				Diesel (L)	LCOE (€/kWh)
	Wind (kW)	PV (kW)	Diesel (kW)	Batteries (kWh)		
Climate 1	100	9.00	35.2	671.1	99,788	0.1760
Climate 2	60	131.50	52.8	534.6	171,480	0.2161
Climate 3	40	157.75	35.2	602.2	156,693	0.2179

consumption is cut by half and, like before, it is limited to the fall and winter months. The energy surplus is rather large, 40%, but the LCOE is at a minimum.

For the second climate scenario, the wind speed is equal to 7.03 m/s and the solar irradiance is 1400 kWh/m^2. The solar PV regains its status in terms of installed capacity but wind is still the main energy source, see Fig. 14. This is actually the only scenario with 3 diesel generators but it is also the one with the lowest storage capacity, mainly because the renewables are more evenly distributed. The energy surplus is 33%.

In the third case, the values for wind speed and solar irradiance are 5.53 m/s and 2200 kWh/m^2. It is the most similar to the reference scenario in terms of installed capacity but the energy surplus is higher (27.3%). The LCOE is roughly the same as in climate

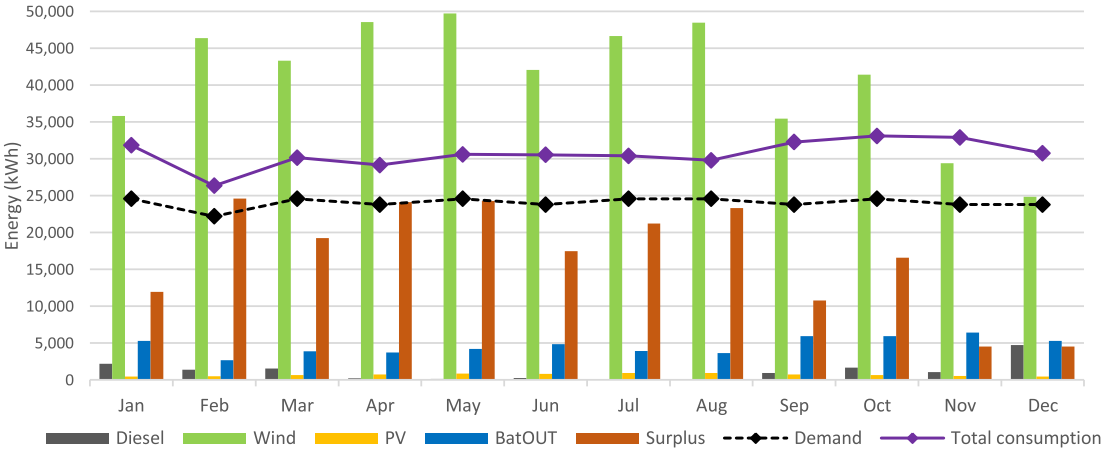


Fig. 13. Monthly energy balance for climate scenario 1.

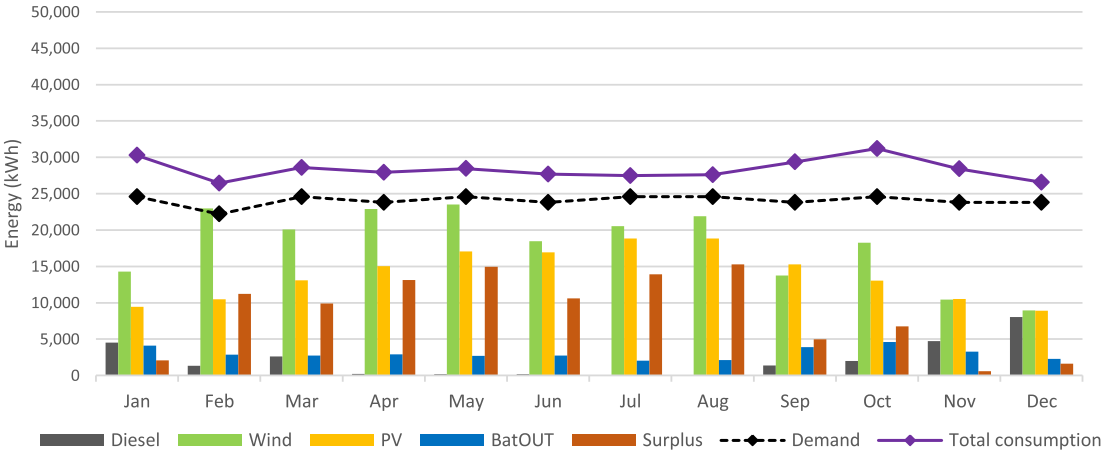


Fig. 14. Monthly energy balance for climate scenario 2.

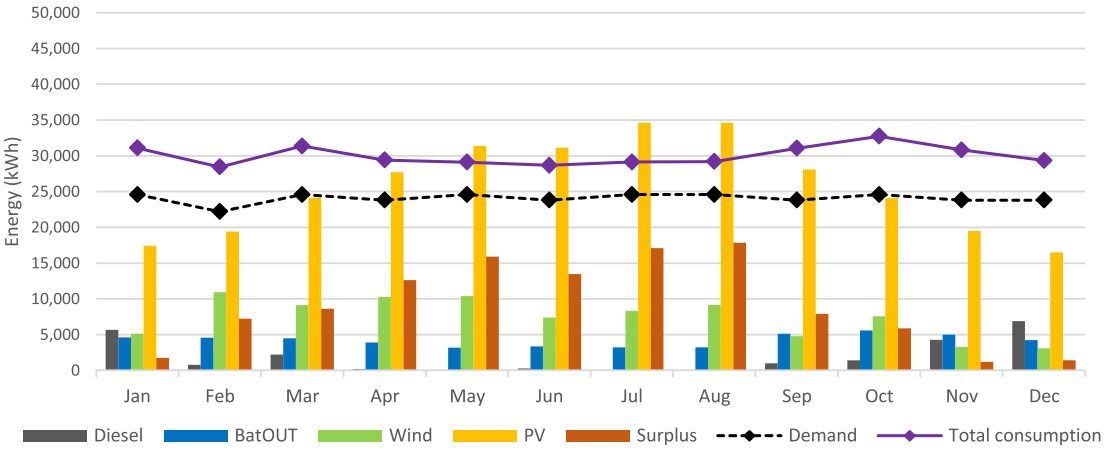


Fig. 15. Monthly energy balance for climate scenario 3.

Table 11
Comparison of sizing results for different demand profiles.

Demand profile	Installed capacity				Diesel (L)	LCOE (€/kWh)
	Wind (kW)	PV (kW)	Diesel (kW)	Batteries (kWh)		
Load A	40	161.5	52.8	558.4	180,844	0.2232
Load B	40	160.75	52.8	662.7	174,086	0.2311

2 but the sizing results are significantly different for all subsystems. The monthly energy balance is shown in Fig. 15.

8.4. Demand profile

Finally, we study the influence of the demand profile on system configuration while keeping the same weekly and annual demand. The new demand profile with two weekly pausing days, labeled Load B in Fig. 4, is typical of most European countries. More specifically, the demand on working days was raised by 18.2%, leading to a lower energy consumption during the weekend. The results in Table 11 show that the total installed power is virtually the same. However, the storage capacity is larger so as to accumulate more energy during the weekend and support the hourly increase in demand during the rest of the week. Driven by the high capital cost of the batteries, and despite the lower diesel consumption, the LCOE increases by 3.5%.

9. Conclusions

This paper has presented a mixed-integer linear programming model for the design of isolated hybrid systems comprising Wind, PV, Batteries and Diesel subsystems. Given the technical specifications of available equipment, meteorological and demand data, the model can determine the installed capacity and the actual number of wind turbines, PV panels and diesel generators of a specific type that are needed, so as to have a more accurate estimation of capital cost. The model solution also provides the optimal hourly operation of each individual subsystem over a long time horizon of one year.

Results for a reference scenario have shown that the optimal configuration features all subsystems, leading to an LCOE of 0.2232 €/kWh and a renewable fraction corresponding to 90.9% of the total energy demand. The PV subsystem had the main share of capacity due to a large correlation between the demand and the daily solar profile. Compared to other configurations with one less subsystem and to a 100% diesel system, important savings were observed. The thorough sensitivity analysis has also shown that the optimal capacity of the individual components is strongly affected by the average wind speed and solar irradiance.

Overall, the developed methodology demonstrated to be flexible and adequate for use as an effective and valuable planning tool for the design of hybrid renewable isolated systems with minimum LCOE during the project lifetime.

Acknowledgments

The authors acknowledge LNEG by providing access to the resources and facilities that enabled to conduct this work. PM Castro gratefully acknowledges financial support from Fundação para a Ciência e Tecnologia (FCT) through the *Investigador FCT 2013* program and project PTDC/EQU-ESI/118253/2010. The research leading to these results has received funding from the European Union Seventh Framework Programme (FP7/2007- 2013) under Grant Agreement N. PCOFUND-GA-2009-246542 and from the Foundation for Science and Technology of Portugal under the Welcome II Programme, Grant DFRH/WIIA/67/2011.

Appendix A. Supplementary data

Supplementary data related to this article can be found at http://dx.doi.org/10.1016/j.renene.2015.04.066.

Nomenclature

Parameters (not provided in the tables)

$a1 - c3$	power curve regression coefficients
C_R	annual replacement cost [€]
CRF	capital recovery factor [%]
E_{con}	annual Energy demand [kWh]
$E_{dem}(t)$	energy demand in time step t [kWh]
$E_{PV}(t)$	energy produced by one PV panel in time step t [kWh]
$E_{WT}(t)$	energy produced by one wind turbine in time step t [kWh]
$G(t)$	solar irradiance at hour t [W/m^2]
I_0	capital cost [€]
$O\&M$	operation and maintenance cost [€]
$T_{amb}(t)$	ambient temperature at hour t [°C]
$T_{cell}(t)$	PV cell temperature at hour t [°C]
$u_z(t)$	wind velocity at height z at hour t [m/s]
$u_{zREF}(t)$	Wind velocity at height reference z_{REF} at hour t [m/s]
z_{REF}	reference height of wind velocity measurement [m]
Δt	time period of 1 h [h]

Variables

C_{bat}	storage capacity to install [kWh]
$C_{dg}(t)$	diesel consumption in time step t [L]
$E_{bat}(t)$	batteries state of charge in time step t [kWh]
$E_{bin}(t)$	available energy for battery charge in time step t [kWh]
$E_{bine}(t)$	energy that effectively is stored in time step t [kWh]
$E_{bout}(t)$	used energy from batteries at time period t [kWh]
$E_{boute}(t)$	energy that effectively is discharged in time step t [kWh]
$E_{dg}(t)$	energy produced by diesel subsystem in time step t [kWh]
$Es_{PV}(t)$	surplus energy from PV subsystem in time step t [kWh]
$Es_{WT}(t)$	energy from wind subsystem in time step t [kWh]
$Eu_{PV}(t)$	used energy from PV subsystem in time step t [kWh]
$Eu_{WT}(t)$	used energy from wind subsystem in time step t [kWh]
$nA(t)$	number of diesel generator starts at hour t
$n_{dg}(t)$	number of diesel generators working at hour t
n_{dgh}	number of annual diesel generators working hours [h]
n_{dgt}	number of diesel generators to install
n_{PV}	number of PV panels to install
n_{WT}	number of wind turbines to install
$TLCC$	total life cycle cost [€]
$TLCC_{bat}$	total life cycle cost of batteries [€]
$TLCC_{dg}$	total life cycle cost of diesel generators [€]
$TLCC_{PV}$	total life cycle cost of PV subsystem [€]
$TLCC_{WT}$	total life cycle cost of wind subsystem [€]
$y(t)$	binary variable for diesel generator subsystem

References

[1] IEA. World energy outlook 2010. 2010.
[2] ONU. United nations development programme. Dezembro 2013 [Online]. Available, http://www.undp.org/content/undp/en/home/ourwork/environmentandenergy/focus_areas/sustainable-energy/universal-access/.
[3] Hunter R, Elliot G. Wind-diesel systems. Cambridge University Press; 1994.
[4] Lundsager P, Baring-Gould EI. Isolated systems with wind power. In: Wind power in power systems. John Wiley & Sons; 2005. p. 299–329.
[5] Zhou W, Lou C, Li Z, Lu L, Yang H. Current status of research on optimus sizing of stand-alone hybrid solar-wind power generation systems. Appl Energy 2010:380–9.
[6] Bajpai P, Dash V. Hybrid renewable energy systems for power generation in stand-alone applications: a review. Renew Sustain Energy Rev 2012;16:2926–39.
[7] Musgrove A. The optimization of hybrid energy conversion systems using the dynamic programming model RAPSODY. Int J Energy Res 1988;12:447–57.
[8] Barley CD, Winn CB. Optimal dispach strategy in remote hybrid power systems. Sol Energy 1996;58:165–79.
[9] Ashok S. Optimised model for community-based. Renew Energy 2007;32:1155–64.
[10] Senjyu T, Hayashi D, Yona A, Urasaki N, Funabashi T. Optimal configuration of power generating systems in isolated island with renewable energy. Renew Energy 2007;32:1917–33.
[11] Vrettos EI, Papathanassiou SA. Operating policy and optimal sizing of a high penetration RES-BESS system for small isolated grids. IEEE Trans Energy Convers September 2011;26(3).
[12] Koutroulis E, Kolokotsa D, Potirakis A, Kalaitzakis K. Methodology for optimal sizing of stand-alone photovoltaic/wind-generator systems using genetic algorithms. Sol Energy 2006;80:1072–88.
[13] Dufo-López R, Bernal-Agustín JL. Multi-objective design of PV-wind-diesel-hydrogen-battery systems. Renew Energy 2008;33:2559–72.
[14] Hafez O, Bhattacharya K. Optimal planning and design of a renewable energy based supply system for microgrids. Renew Energy 2012;45:7–15.
[15] Morais H, Kádár P, Faria P, Vale ZA, Khodr HM. Optimal scheduling of a renewable micro-grid in an isolated load area using mixed-integer linear programming. Renew Energy 2010;35(1):151–6.
[16] Palma-Behnke R, Benavides C, Lanas F, Severino B, Reyes L, Llanos J, et al. A microgrid energy management system based on the rolling horizon strategy. IEEE Trans Smart Grid 2013;4(2).
[17] Bixby R, Rothberg E. Progress in computational mixed integer programming - a look back from the other side of the tipping point. Ann Operations Res 2007;49(1):37–41.
[18] Lima RM, Grossmann IE. Computational advances in solving mixed integer linear programming problems. Chem Eng Greetings Prof. Sauro Pierucci, AIDIC 2011:151–60.
[19] Aeolos, "Aeolos wind turbine," [Online]. Available: http://www.windturbinestar.com/english.html. [Accessed November 2013].
[20] C&F Green Energy, "C&F Green Energy," [Online]. Available: http://www.cfgreenenergy.com/. [Accessed November 2013].
[21] Polaris, "Polaris," [Online]. Available: http://www.polarisamerica.com/. [Accessed November 2013].
[22] Lorenzo E. Energy collected and delivered by PV modules. In: Handbook of photovoltaic science an engineering. John Wiley & Sons; 2011. p. 984–1042.
[23] Aksa, "Aksa Power Generation," [Online]. Available: http://www.aksa.com.tr/en/main-page. [Accessed May 2013].
[24] FG Wilson, [Online]. Available: https://www.fgwilson.com/. [Accessed May 2013].
[25] Olympian, [Online]. Available: http://www.olympianpower.com/. [Accessed May 2013].
[26] Diaf S, Notton G, Belhamel M, Haddadi M, Louche A. Design and techno-economical optimization for hybrid PV/wind system under various meteorological conditions. Appl Energy 2008b;85:968–87.
[27] Short W, Packey DJ, Holt T. A manual for economic evaluation of energy efficiency and renewable energy technologies. NREL 1995. NREL/TP-462-5173, http://www.nrel.gov/docs/legosti/old/5173.pdf.
[28] IEA&NEA. Projected costs of generating electricity 2010. 2010.
[29] Sandia National Laboratories. Electricity storage handbook. 2013.
[30] LNEG. Análise de desempenho térmico, económico e ambiental de sistemas solares – SolTerm5. 2013 [Online]. Available, http://www.lneg.pt/iedt/projectos/370/resumo.